



9.7.5 Wrap-Up: Cool Down

1. Provide a real-world context that would require you to find the volume of a three-dimensional figure.



Unit 9, Lesson 8: Surface Area of Three-Dimensional Figures



Benchmark

MA.912.GR.4.6 Solve mathematical and real-world problems involving the surface area of three-dimensional figures limited to cylinders, pyramids, prisms, cones, and spheres.



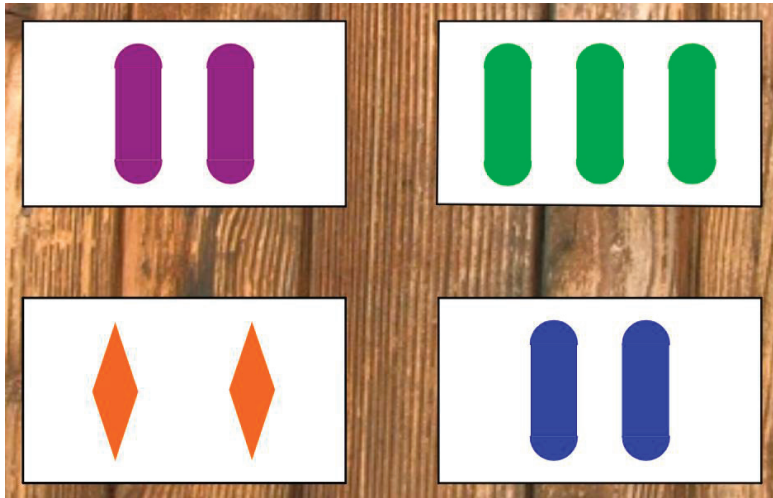
Learning Targets

- ☐ I can solve mathematical problems involving the surface area of cylinders, pyramids, prisms, cones, and spheres.
- ☐ I can solve mathematical problems involving the lateral area of cylinders, pyramids, prisms, cones, and spheres.



9.8.1 Warm-Up: Which One Doesn't Belong?

1. Which one doesn't belong? Explain your answer.



9.8.2 Collaboration: Using Nets to Find Surface Area



The **lateral area** of a three-dimensional figure is the sum of the area of the figure's faces.



The **surface area** of a three-dimensional figure is the sum of the lateral area of the figure's faces and the area of the figure's bases.

1. Work with your partner to determine a formula for the lateral area and the surface area of the figure you have been given. Complete the table.

Our figure was:				
Dimension				
Variable Used				



9.8.3 Activity: Gallery Walk

- During the gallery walk, record the formulas. Make any useful notes on each figure's net.

Figure	Net
Cylinder	
Lateral Area =	
Surface Area =	
Cube	
Lateral Area =	
Surface Area =	
Rectangular Prism	
Lateral Area =	
Surface Area =	

Figure	Net
Triangular Prism	
Lateral Area =	
Surface Area =	
Square Pyramid	The height of the triangle that is a face of a pyramid is also known as slant height, h_s.
Lateral Area =	
Surface Area =	
Rectangular Pyramid	The height of the triangle that is a face of a pyramid is also known as slant height, h_s.
Lateral Area =	
Surface Area =	



9.8.4 Bring It Together: Formulas

Using the net of three-dimensional objects can help to find the lateral area and surface area. The net of a cone and a sphere make deriving the formulas for surface area from the net more challenging. A chart of formulas is given, where SA represents surface area.

Figure	Surface Area	Variables
Prism	$SA = 2B + Ph$	B = base, P = perimeter, h = height
Cylinder	$SA = 2B + Ch$ Or $SA = 2B + Ph$	B = base, C = circumference (or P = perimeter, h = height)
Cone	$SA = B + \pi r h_s$	B = base, r = radius, h_s = slant height
Regular Pyramid (a pyramid whose base is a regular polygon)	$SA = B + \frac{1}{2}Ph_s$	B = base, P = perimeter, h_s = slant height
Sphere	$SA = 4\pi r^2$	r = radius

1. The formulas in the table may not match the formulas recorded during the gallery walk. Which figures have formulas that are different?

2. There are many ways to write and to understand formulas. Discuss with your partner which formulas make more sense to you. Discuss how the formulas in the formula chart can be used to help you remember how to find surface area. Summarize your discussion.

3. The chart does not have a formula for lateral area. For each of the figures, except for spheres, circle the part of the surface area formula that gives the lateral area.



9.8.5 Guided Instruction: Lateral Area and Surface Area

1. A square pyramid has a slant height of 7 inches, and the area of the base is 16 square inches. What is the surface area, in square inches?
2. The surface area of a sphere is 144π feet. What is the radius of the sphere?
3. If a cylinder has a height of 16 centimeters and a radius of 8 centimeters.
 - a. Find the lateral area. Leave your answer in π form.
 - b. Find the surface area. Leave your answer in π form.
4. A right triangular prism has base side lengths of 8, 15, and 17 feet (ft), where 17 ft is the hypotenuse of the triangle. The prism has a height of 7 ft. What is the surface area, in square feet, of the triangular prism?





9.8.6 Guided Practice: Lateral Area and Surface Area

1. A pyramid has a square base whose area is 49 square feet with a height of 11.5 feet. Find the lateral area, in square feet, of the pyramid.
2. A rectangular prism has a length of 12.5 inches (in.), a width of 8 in., and a height of 5 in. Determine the surface area, in square inches, of the rectangular prism.
3. The radius of a sphere is 18.7 feet. Find, in square feet, the surface area of the sphere. Round your answer to thousandths place.



9.8.7 Your Turn!

1. Find the lateral area, in square meters, of a cone that has a radius of 3 meters (m) and a slant height of 8.5 m. Round your answer to thousandths place.

2. A cylinder has a radius of 5 inches (in.) and a height of 3.75 in. Find the surface area, in inches, of the cylinder. Round your answer to thousandths place.



9.8.8 Wrap-Up: Error Analysis

1. Cody and Mila are finding the surface area of a pyramid whose base area is 144 inches and slant height is 16 inches. Their work is shown in the table.

Cody's Work	Mila's Work
$SA = B + \frac{1}{2}Ph_s$	$SA = B + \frac{1}{2}Ph_s$
$SA = 144 + \frac{1}{2}(48)(16)$	$SA = (144) + \frac{1}{2}(144)(16)$
$SA = 528$ square inches	$SA = 1296$ square inches

- a. Whose work is correct?

- b. Write a note to the student whose work is incorrect describing the mistake the student made.